FORM 2

THE PATENTS ACT, 1970 (39 of 1970)
& THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. TITLE OF THE INVENTION:

A System for Detecting and Preventing Unauthorized Access to Digital Documents Using Adaptive Access Control

2. APPLICANT(S):

Applicant information to be designated upon official lodgment.

3. ABSTRACT

A computer-implemented document security system comprising an access request processor, a contextual risk assessment module, and an adaptive access control engine configured to evaluate access conditions and modify document access permissions in response to detected security risks.

4. FIELD OF THE INVENTION & BACKGROUND

2. Background of the Invention

The present invention relates to the field of cybersecurity and digital document access management, particularly systems and methods for controlling access to sensitive electronic documents based on user permissions and contextual security conditions.

Existing document management systems employ access control mechanisms such as role-based access control, identity verification, multi-factor authentication, and predefined permission policies to regulate access to digital information. These mechanisms help protect confidential documents by restricting access to authorized users and maintaining records of document-related activities. Some advanced systems also incorporate behavioral monitoring and risk-based authentication to identify potentially suspicious access attempts.

However, conventional access management approaches may rely substantially on predefined permissions and authentication decisions made at the time of access. Changes in user behavior, device conditions, access context, or other security indicators may not always result in timely adjustments to existing document permissions. Furthermore, security monitoring and access enforcement may operate as separate processes, potentially delaying responses to suspicious activities.

These limitations highlight the technical need for an integrated document access management mechanism capable of evaluating relevant security parameters, identifying potentially unauthorized access, and applying appropriate access restrictions according to the assessed risk. Such a mechanism may improve the responsiveness, consistency, and traceability of document security operations while supporting controlled access to sensitive information.

The invention is proposed in this technical context and aims to address the identified limitations through an appropriate combination of contextual risk assessment and adaptive access control mechanisms. The specific technical features and their potential advantages are described in the subsequent sections of this disclosure.

5. TECHNICAL DEFECTS & PROBLEM STATEMENT

Existing document-sharing systems may allow unauthorized users to access sensitive information when access permissions are improperly configured or user credentials are compromised. Conventional access control mechanisms may not respond dynamically to suspicious access patterns, increasing the risk of data leakage.

6. PROPOSED TECHNICAL SOLUTION

The proposed system monitors document access requests and evaluates user permissions, device information, and access patterns. A risk assessment module identifies potentially unauthorized activity and dynamically applies access restrictions, additional verification, or access denial based on configurable security rules.

7. SUMMARY OF THE INVENTION

5. Summary of the Invention

The present invention relates to a system and method for adaptive access control of digital documents using contextual security assessment. The invention is intended to address limitations associated with conventional document access management systems that rely primarily on predefined permissions and static authentication decisions.

In one embodiment, the system comprises a document access request processing module, a user identity and permission verification module, a contextual risk assessment module, an access control decision engine, and an audit logging module. The system evaluates an access request using relevant parameters, which may include user permissions, device information, authentication status, access history, and detected behavioral patterns. Based on the evaluated conditions, the system determines whether access should be granted, restricted, subjected to additional verification, or denied.

In another embodiment, the system dynamically reassesses access conditions during a document access session or when significant security-related events occur. The access control decision engine may update applicable permissions or initiate additional security measures when the assessed risk changes. The resulting decisions and associated security events may be recorded to support traceability, monitoring, and subsequent review.

In an alternative embodiment, the system may be implemented as a web-based application, an enterprise document management component, a cloud-based security service, or an application programming interface integrated with an existing document repository. The contextual risk assessment may employ rule-based evaluation, statistical analysis, or machine-learning techniques, depending on the implementation requirements.

The proposed system aims to provide a coordinated mechanism for contextual risk evaluation and access enforcement, thereby improving the responsiveness and consistency of digital document protection. The particular technical implementation, operational parameters, and combination of features are to be defined in the detailed description and claims of the invention.

8. DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

6. Detailed Description

The present invention relates to a system and method for adaptive access control of digital documents using contextual security assessment. The system is configured to evaluate document access requests, determine applicable access permissions, assess security-related conditions, and execute access control decisions based on the evaluation.

6.1 System Architecture

In one embodiment, the system comprises a user interface (100), an access request processor (110), an identity and permission verification module (120), a contextual data collection module (130), a risk assessment engine (140), an access control decision engine (150), a document access enforcement module (160), a document repository (170), and an audit logging module (180). The modules communicate through suitable software interfaces and may be deployed on a single computing device, a distributed computing environment, or a cloud-based infrastructure.

The user interface (100) enables a user to select a document and submit an access request. The access request

9. TECHNICAL COMPONENTS & ASSEMBLIES

7. Technical Components

The following reference numerals identify the principal components of the illustrative system.

Reference numeral	Component	Function
100	User Interface	Receives document access requests and displays access outcomes.
110	Access Request Processor	Receives and processes access request information.
120	Identity and Permission Verification Module	Verifies authentication status and retrieves applicable permissions.
130	Contextual Data Collection Module	Collects relevant security and access-context information.
140	Risk Assessment Engine	Evaluates contextual information and generates a risk assessment.
150	Access Control Decision Engine	Determines the access action according to risk and policy.
160	Document Access Enforcement Module	Enforces the access decision.
170	Document Repository	Stores or provides access to protected digital documents.
180	Audit Logging Module	Records access events, assessment results, and decisions.

Computational and Communication Elements

The system may additionally comprise one or more processors, memory devices, persistent storage, network interfaces, and communication channels configured to execute the software modules and exchange information between them.

The processors execute instructions stored in memory. The memory stores program instructions, configuration parameters, access policies, and temporary processing data. Persistent storage may maintain document metadata, access records, and audit logs. Communication interfaces may enable interaction with external authentication providers, document repositories, and enterprise identity services.

These supporting computing elements are implementation-dependent and should be assigned separate reference numerals in the final drawings if they are important to understanding the claimed invention.

10. WORKING PRINCIPLE & OPERATIONAL FLOW

8. Working Principle

The system operates according to the following sequence.

Step 1: Access Request Initiation

A user selects a digital document through the user interface (100) and requests an operation, such as viewing, downloading, or modifying the document.

Step 2: Request Processing

The access request processor (110) receives the request and identifies the requesting user, the target document, the requested operation, and the available session context.

Step 3: Identity and Permission Verification

The identity and permission verification module (120) checks the user's authentication status and retrieves the permissions applicable to the requested document operation. A request that fails mandatory authentication or authorization requirements may be rejected according to the applicable policy.

Step 4: Contextual Information Collection

For requests that proceed to risk assessment, the contextual data collection module (130) obtains available security indicators, such as device information, access history, authentication status, and request patterns, subject to the system's permissions and privacy requirements.

Step 5: Risk Assessment

11. TECHNICAL ADVANTAGES & IMPROVEMENTS

The proposed invention may combine contextual risk assessment with a dynamic access enforcement mechanism that adjusts document permissions in response to changing security conditions. A potential inventive contribution would be a specific technical mechanism for coordinating these components, such as a defined risk-update process, a novel policy-adjustment algorithm, or a measurable improvement in access-control performance. The inventive-step argument must identify the distinguishing technical features, explain why they are not disclosed by the closest prior art, and establish why a person skilled in the field would not have arrived at the claimed combination in an obvious manner.

12. APPLICATIONS & EMBODIMENTS

10. Applications

The proposed system for adaptive access control of digital documents may be applied in environments where electronic documents contain confidential, sensitive, proprietary, or restricted information. The system can be integrated into existing document management infrastructure to evaluate access requests and enforce security policies based on user permissions and relevant contextual information.

10.1 Enterprise Document Management: The system may be used by organizations to manage access to internal reports, business records, contracts, technical documentation, and other confidential corporate documents.

10.2 Healthcare Information Systems: The system may be adapted to protect electronic medical records, diagnostic reports, and administrative healthcare documents by enforcing role-based permissions and additional security checks where appropriate.

10.3 Educational and Research Institutions: Universities, research laboratories, and academic institutions may use the system to control access to research papers, unpublished findings, intellectual property disclosures, student records, and institutional documents.

10.4 Financial and Banking Environments: The system may be incorporated into document management workflows involving financial reports, audit records, customer documentation, and internal compliance materials, subject to applicable regulatory requirements.

10.5 Government and Public Administration: Government departments and public institutions may use the system to manage access to administrative records and restricted digital documents in accordance with applicable information security policies.

10.6 Cloud-Based Collaboration Platforms: The system may be integrated with cloud storage services, collaborative workspaces, and enterprise content management applications to support contextual access evaluation across distributed working environments.

10.7 Intellectual Property and Legal Services: Patent professionals, legal firms, research organizations, and innovation teams may use the system to control access to patent drafts, invention disclosures, confidential technical documents, and supporting evidence.

The system may be deployed as a standalone application, a server-side software service, a cloud-hosted security component, or an application programming interface integrated with an existing document repository. The precise implementation may be adapted according to the security requirements, scale, and operational environment of the intended application.

13. INDUSTRIAL APPLICABILITY

11. Industrial Applicability

The invention relates to the technical field of cybersecurity and digital document access management and is capable of practical implementation in information technology systems used by commercial, industrial, educational, healthcare, financial, legal, and public-sector organizations.

The system can be implemented using conventional computing hardware, including one or more processors, memory devices, persistent storage, and network communication interfaces, executing software instructions that implement contextual risk assessment and access control functions. It may be manufactured or configured as a software product, deployed as a server-based application, incorporated into an enterprise document management platform, or provided as a cloud-based service.

In industrial use, the system receives document access requests, evaluates relevant authentication and contextual information, determines an access decision according to configured policies, and enforces the resulting decision. The system may also maintain audit records to support security monitoring and operational review.

The implementation can be adapted to different document repositories, organizational access policies, and computing environments. Its use may assist organizations in managing access to confidential information, applying security controls consistently, and maintaining records of document access events.

Accordingly, the described system is capable of practical use in information technology and digital information security operations. The precise industrial applicability and technical advantages depend on the final implementation and the features supported by the invention.

This section is intended to address the industrial applicability requirement described in Section 2(1)(ac) of the Indian Patents Act, 1970, subject to the actual technical disclosure and applicable patent examination requirements.